Camera and Directory — QA Test Cases

Task 6.12 · v0.4 (draft) · Owner: Syed Sarim Jawad Zaidi · Section C1 from docs/api/directory_contract.md (Task 6.1, PR #60), C2 from docs/api/booking_contract.md (Task 6.2, PR #67); A and B from docs/api/camera_contract.md (Tasks 6.3, 6.4); SRS v1.0 FR-K1, FR-K2, FR-N1, FR-N3, FR-O1 and the 6.11 mockups (18–18e, 19–19d, 29, 29b v0.2)

Purpose: above all, prove that the Dependent ALWAYS sees a notice while a Guardian watches the camera (FR-K1.3, a privacy requirement); then that clips are saved on Panic and Remote alarm, and that the directory and home-care requests work. Results go on a copy of the QA Checklist template.

Status: the directory (C1) follows the 6.1 contract and booking + activity log (C2) the 6.2 contract. The camera (A, B) follows the 6.3 / 6.4 contract; Talha's 6.9 camera screen is built without real streaming, so the live parts of A and B wait for 6.10 (the spare phone) and a real ZegoCloud room.

1. Test setup

1.1 Test accounts

Role	Name in tests	Set-up
Dependent	Ammi	Registered, KYC verified, linked to Ali.
Primary Guardian	Ali	Phone + CNIC verified.
Secondary A	Sara	Invited by Ali, verified. Permission Camera viewing = ON.
Secondary B	Bilal	Invited by Ali, verified. Camera viewing OFF.
Outsider	Zain	A verified Guardian NOT linked to Ammi (for refusal checks).

1.2 How to run

- **Devices:** the spare phone in camera mode in the room (6.10), Ammi's phone, Ali's phone, Sara's phone.
- **Order:** after Abdullah's 6.1–6.4 and Talha's 6.8–6.9.
- **Privacy rule:** K-02 to K-04 are run 10 times each. A single miss is a Critical bug and blocks ICP-6.

2. Test cases

A. Camera live view and the viewing notice (FR-K1, Task 6.3)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
K-01	FR-K1	Room phone paired (signed in as Ammi, camera mode), online.	Ali → Camera (19).	Live video starts (the phone publishes only now); LIVE badge, timer, "2 of 3 watching" when Sara is also in. Every 30 s the app renews quietly; watching does not stop by itself.		
K-02	FR-K1.3	Ammi's phone on any screen.	Ali starts viewing. Repeat with Ammi on: Home, Medicine, Wellness prompt, Messages, phone locked, another app open.	Ammi sees the viewing notice with Ali's name on EVERY screen (29), 10 / 10 times, and gets a camera_view notice; the room phone shows who is watching (29b).		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
K-03	FR-K1.3	Ali watching.	Ali taps Stop viewing; separately, force-close Ali's app; separately, turn Ali's internet off.	Stop: the notice goes at once. Closed app / no internet: the session ends within 90 s (no renewal) and the notice goes. It never stays on by mistake.		
K-04	FR-K1.3	Ali and Sara watching.	Both open the camera; Ali stops.	Notice names both ("Ali and Sara are watching"); after Ali stops it shows only Sara.		
K-05	FR-K1	Ali, Sara and a third Guardian watching.	A 4th try (Postman with another linked Guardian if available, else read the contract limit).	409 TOO_MANY_VIEWERS; app shows "Camera busy, try again shortly" (19d note).		
K-06	FR-A4	Bilal camera_viewing OFF.	Bilal looks for Camera; Postman GET …/camera/stream as Bilal.	Camera hidden or read-only note for Bilal; 403 MODULE_ACCESS_DENIED.		
K-07	FR-A4	Sara watching.	Ali switches Sara's camera_viewing OFF.	Sara is cut off at her next renewal at the latest (within 30 s); the notice on Ammi's phone updates.		
K-08	FR-K1	Room phone switched off or offline.	Ali opens Camera.	409 CAMERA_OFFLINE: "Camera is offline" (19c) with the last seen time; Try again works when the phone is back.		
K-09	FR-K1	No camera paired.	Ali opens Camera; then Ali (Primary) / Sara (Secondary) try to pair in Postman.	409 NO_CAMERA: "No camera set up yet" (19d). Only Ammi or the Primary can pair (Sara: 403 FORBIDDEN).		
K-10	FR-N3	K-01 – K-07 done.	Ali opens the audit log (31).	camera_view_started / ended entries with who and when; only the Primary can open the log.		
K-11	FR-N1	—	Check the stream answer in Postman and the security statement (docs/backend/security_encryption.md).	Token responses carry Cache-Control: private, no-store; video is encrypted in transit by the provider (not end-to-end, as stated).		

B. Emergency clips (FR-K2, FR-O1, Task 6.4)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
K-12	FR-K2	Room phone paired and online.	Ammi presses Panic.	The room phone records 20 s (29b shows it) and uploads; the clip appears in Emergency clips (19b) as Panic · Saved; Ali and Sara (camera_viewing) get a camera_clip notice.		
K-13	FR-K2	Room phone paired and online.	Ali rings the Remote alarm.	Clip saved, labelled Remote alarm.		
K-14	FR-K2	Room phone offline.	Ammi presses Panic; wait 3 minutes.	The clip shows as Missed (no video). Panic itself works as normal.		
K-15	FR-A4, FR-N3	Bilal camera_viewing OFF.	Bilal tries to list or open a clip; Ali downloads one.	Bilal refused (403); Ali's download is in the audit log (camera_clip_viewed).		
K-16	FR-K2	Clips exist.	Sara tries to delete a clip; Ali (Primary) deletes one.	Sara refused; Ali's delete removes it (204), audit log camera_clip_deleted.		
K-17	FR-O1	—	Read the retention rule (docs/backend/data_retention.md).	Clips kept while the account is active, removed with the account after the 60-day closure period.		

C1. Directory — read side (FR-I1, FR-J1, Task 6.1)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
D-01	FR-I1	Any signed-in user (Ali, Ammi or an unverified Guardian).	Directory → Emergency help (18).	List loads for everyone; filter chips come from /organizations/types (hospital, clinic, pharmacy, ambulance, emergency service, blood bank, laboratory).		
D-02	FR-I1	Location allowed.	Open 18 with "Near you".	Entries with a position show distance (e.g. 2.4 km), nearest first; the national numbers (1122, 115, 1020, 1021, 15, 16) always stay in the list, after the located ones, with no distance.		
D-03	FR-I1	Location NOT allowed.	Open 18.	List sorted by name; no distances; no position is sent.		
D-04	FR-I1	—	Search "blood", "Lahore", "%"; filter Pharmacy; Open 24 h only; a radius with no position (Postman).	Search matches name, city, address, services and type; % is a normal letter; filters work; radius without lat/lng: 422 VALIDATION_ERROR.		
D-05	FR-I1	—	Tap Call on Rescue 1122; then look at the Call button of a sample entry and of an entry with no phone.	1122 opens the phone dialer with 1122. Sample entries show the Sample label and cannot be dialled as real; an entry with no phone has no Call button.		
D-06	FR-J1	—	Home care (18b): filter Nurse; sort Lowest charge; open a provider with no charge.	Chips come from /providers/specializations; sort works (unknown charge last); charges in PKR with the unit (visit / hour / day / month); no charge = "Ask for price".		
D-07	FR-J1	—	Postman: min_charge above max_charge; sort "fast"; limit 101; an unknown provider id.	422 VALIDATION_ERROR for the first three; 404 NOT_FOUND for the unknown id.		
D-08	FR-P1	Ammi logged in.	Find the quick emergency numbers in Dependent Mode.	Ambulance numbers available without a position (from ? type=ambulance).		

C2. Home-care requests and Activity Timeline (FR-J1, FR-I2/J2, Task 6.2)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
D-09	FR-J1	Ali (Primary) logged in.	Home care → a provider → Request (18c): wished time Fri 10:00, note → Save request.	201; request shows as Pending (18e) with the provider's name, charge in PKR and phone + Call; nothing is sent to the provider.		
D-10	FR-J1	—	Save with no wished time; with a time in the past; more than a year ahead; a note of 301 characters.	No time is allowed (optional). Past / over a year / long note: 422 VALIDATION_ERROR, nothing saved.		
D-11	FR-J1	D-09 request open.	Request the same provider again for Ammi.	409 BOOKING_ALREADY_OPEN; the app says a request is already open.		
D-12	FR-J1	Weak internet.	Tap Save, then tap again / retry (same client_id).	Only one request saved; the retry returns the first one (already_saved: true).		
D-13	FR-J1	D-09 request Pending.	Ali: Mark as confirmed → Completed. A second request: cancel it. Then try to confirm the completed one (Postman).	Pending → Confirmed → Completed; Cancelled from Pending; finished requests cannot change: 409 BOOKING_STATUS_CONFLICT.		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
D-14	FR-J1, FR-A4	Sara (health_records ON) made a request; Bilal OFF.	Sara confirms her own request; Sara tries to change Ali's request; Bilal opens Home care requests.	Sara can change her own; Ali's: 403 FORBIDDEN; Bilal: 403 MODULE_ACCESS_DENIED (no list).		
D-15	FR-I2	—	Tap Call on City Care Pharmacy twice within a minute; then once more after a minute.	Activity log shows "Called City Care Pharmacy" once for the first two taps, then a second entry.		
D-16	FR-I2/J2	Calls and requests by Ammi, Ali and Sara.	Ali, Sara and Ammi each open the Activity log (18d); filter Calls, then Requests.	One timeline, newest first, the same for everyone; each line says who (name and role); request lines show the current status; filters work.		
D-17	FR-M1	Sara has health_records.	Ali makes a request; then Ali confirms it.	Sara and Ammi get a standard booking_update notice each time; Ali (who acted) does not.		
D-18	FR-J3	—	Look through the request flow for any payment step.	No payment anywhere (future scope).		

3. Open points

- Timings come from the contract: renewal every 30 s, a session ends 90 s after the last renewal, clips are 20 s and expire after 3 minutes, at most 3 viewers. The notice appearing within a few seconds depends on how often the Dependent app polls; agree that with Talha before ICP-6.
- Directory content: only the 6 national emergency numbers are real; all other entries are samples to be replaced before the final demo (clarification 121, who collects them is open).
- Open with the team (clarifications 128, 129): the family keeps the request status themselves, and health_records is the permission for requests and the timeline; confirm both before ICP-6.